MALSHA NETHMINI

DATA SCIENCE UNDERGRADUATE

Tissamaharama, Sri Lanka | nsirimanna25@gmail.com | +94 72 971 0871 | LinkedIn | GitHub

PROFILE

Third-year BSc (Hons) Data Science undergraduate with hands-on experience in Python, SQL, machine learning, exploratory data analysis, statistical analysis, and model deployment. Experienced in building end-to-end data science projects covering data cleaning, feature preparation, model training, evaluation, and interactive application development using Scikit-learn, PyTorch, and Streamlit. Interested in applying data-driven problem solving and analytical skills to real-world business problems through a Data Science or Data Analytics internship.

TECHNICAL SKILLS

Programming: Python, SQL
Data Science: Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Data Visualization
Machine Learning: Regression, Classification, Clustering, Model Evaluation
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, torchvision, Matplotlib, Streamlit
Databases: MySQL, MongoDB
Tools: Git, GitHub, Jupyter Notebook, Visual Studio Code
Web & Deployment: HTML, CSS, JavaScript, PHP, Streamlit Community Cloud

PROJECTS

Heart Disease Risk Prediction

Python | Scikit-learn | Pandas | Streamlit

- Developed an end-to-end machine learning application to predict heart disease risk using clinical attributes including chest pain type, cholesterol, blood pressure, and ECG results.
- Performed exploratory data analysis and data preprocessing, identifying and removing 723 duplicate records from 1,025 observations, resulting in 302 unique patient records and reducing potential data leakage.
- Trained and compared Logistic Regression, Decision Tree, and Random Forest classification models.
- Selected Random Forest as the final model, achieving 83.6% accuracy and 83.9% F1-score.
- Developed and deployed an interactive Streamlit application with guided patient-data entry, real-time risk predictions, a visual risk gauge, and feature-importance analysis.

FairVision: Bias Detection and Mitigation in a CNN Age Classifier

PyTorch | Hugging Face Datasets | scikit-learn | Streamlit

- Built a CNN from scratch in PyTorch (no pretrained models) to classify 9 age groups on the FairFace dataset using a 20,000-image stratified subset, reaching 43.2% baseline test accuracy.
- Audited fairness across race/gender, finding a 14.3-point race accuracy gap; applied class-weighted loss and oversampling, cutting the gap to 6.3 points and the gender gap to 0.3 points (macro F1 0.29 → 0.36).
- Deployed a Streamlit web app with face cropping, top-3 predictions and low-confidence warnings.

Sri Lankan Property Price Predictor

Python | Scikit-learn | FastAPI | Streamlit | Railway

- Built an end-to-end ML web app predicting property prices across Sri Lanka from 149,635 cleaned real listings (200K+ raw ikman.lk ads, Kaggle).
- Engineered features from messy pricing formats and nested JSON fields, including converting per-perch land prices to total price.
- Trained and compared Linear Regression and Random Forest models, achieving R^2 of 0.81 (MAE ~Rs 5.26M) with Random Forest.
- Built a REST API with FastAPI and a Streamlit frontend, deployed live on Railway and Streamlit Community Cloud.

EDUCATION

BSc (Hons) in Data Science | **Sabaragamuwa University of Sri Lanka** **2024 - Present**

CERTIFICATIONS

Certified AI & ML Engineer (CAME) | **Institute of Software Engineering (IJSE), Sri Lanka** **Nov 2025 - Apr 2026**
Six-month programme covering: Python for Data Science | Machine Learning | Deep Learning | Natural Language Processing | Large Language Models | MLOps.

SOFT SKILLS

Problem Solving | Analytical Thinking | Teamwork | Communication | Research | Presentation